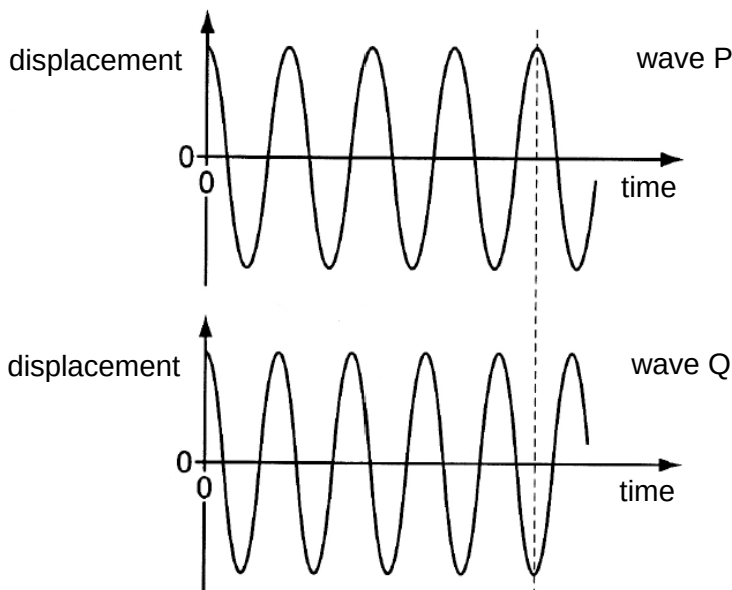


- 16 The diagram shows the displacement-time graphs of two sound waves P and Q at a point in space. The graphs have the same scales for the time axes.



The frequency of Q is 125 Hz. The waves are in phase at time = 0.

At what time are the waves next in phase?

- A 32 ms B 36 ms C 64 ms D 72 ms

Ans: D

From graphs shown,

P and Q are initially in phase at $t = 0$, and become out of phase next at $t = 4.5T_Q$, where T_Q is the period of wave Q.

$$T_Q = 1/f_Q = 1/125 \text{ s}$$

Hence the next time the waves become in phase must be for another duration of $4.5 T_Q$,

i.e. they are next in phase at time

$$t = 4.5 T_Q + 4.5 T_Q = 9 T_Q = 0.072 \text{ s}$$

- 17 Which of the following correctly states the changes, if any, in the potential energy and kinetic